PROJECT DOCUMENTATION REPORT

Visualizing Housing Market Trends

Project Title	Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization
Repository	https://github.com/AbhinavSaxena-spec/Strategic-Product-Placement-Analysis-Unveiling-Sales-Impact-with-Tableau-Visualization.git
Tech Stack	Tableau, Python (Flask), JavaScript, CSS, HTML
Architecture	Frontend / Backend Separated
Report Date	March 7, 2026
Status	Completed

1. Executive Summary

This document provides comprehensive project documentation for the Strategic Product Placement Analysis project. The project has been restructured with a clean separation of frontend and backend concerns, delivering an improved, modular architecture compared to previous iterations.

Using Tableau for data visualization and a Python/Flask backend paired with a JavaScript-rich frontend, the solution enables stakeholders to explore and interpret product placement dynamics through interactive dashboards embedded in a professional web application.

2. Project Overview

2.1 Project Information

Project Name	Strategic Product Placement Analysis
Repository URL	https://github.com/AbhinavSaxena-spec/Strategic-Product-Placement-Analysis-Unveiling-Sales-Impact-with-Tableau-Visualization.git
Primary Analytics Tool	Tableau (Product_Placement_Analysis.twb)
Backend Framework	Python / Flask
Frontend	JavaScript, CSS, HTML
Architecture Pattern	Frontend / Backend Separated
Version Control	Git / GitHub

2.2 Project Objectives

The project addresses the following business challenges:

- **Measure Sales:** Track how placement impacts revenue.
- **Analyze Consumers:** Map demographics to purchasing behavior.
- **Build Dashboards:** Visualize data using Tableau and Flask.
- **Optimize Positioning:** Identify the most effective placement locations.
- **Maximize ROI:** Drive data-backed decisions to boost profits.

2.3 Key Stakeholders

Film & TV Producers	Maximize brand partnership revenue.
Retailers & Consumer Goods	Boost in-store and online sales..
Advertising Agencies	Improve client campaign ROI.

3. Technical Architecture

3.1 Architecture Overview

The new version of the project adopts a clean frontend/backend separation, improving maintainability, scalability, and developer experience. The system consists of three primary layers:

- **Data Layer:** Processes and structures raw sales, consumer demographic, and product placement data.
- **Visualization Layer (Tableau):** Transforms the processed data into interactive dashboards and analytical charts.
- **Application Layer (Flask):** Acts as the web framework to securely embed and serve these dashboards to end-users.

3.2 Language & Technology Breakdown

The following table reflects the actual language composition of the repository as reported by GitHub:

Language / Technology	Usage %	Role in Project
CSS	**54.7%**	Styling and visual design of the frontend web application.
JavaScript	**40.9%**	Frontend interactivity, Tableau embed handling, and dynamic UI components.
Python	**2.9%**	Flask backend — routing, API endpoints, and server-side logic.
HTML	**1.5%**	Base page structure and Jinja2 template markup.

3.3 Full Technology Stack

Data Visualization	Tableau Desktop / Tableau Public
Backend	Python 3 with Flask web framework
Frontend Scripting	JavaScript
Frontend Styling	CSS3

Markup	HTML5
Version Control	Git / GitHub
Deployment	Tableau Public (embed) + Flask server

4. Data Overview

4.1 Dataset Description

The project is powered by a pre-processed placement dataset (Product Positioning.csv) containing cleaned and transformed product positioning records. Key metrics derived from the data include:

- **Sales Figures:** Revenue, units sold, conversion rates, and overall ROI.
- **Placement Metrics:** Specific physical locations (store aisles/shelves), digital spots (website banners), and media timestamps (movie scenes/TV episodes).
- **Consumer Demographics:** Customer age, gender, location, and purchasing behavior.
- **Engagement Data:** High-traffic area tracking, web clicks, audience viewership, and interaction rates.

4.2 Data Preparation Steps

- Collection of raw product position data
- Connection of the dataset to Tableau for initial profiling
- Data type validation and field transformation
- Creation of calculated fields
- Filter and aggregation configuration for optimal dashboard performance.

5. Visualization Scenarios

Four core visualization scenarios were developed within the Tableau workbook to address project objectives:

Scenario 1: Overall Data Overview

Visualization Type: Summary

Presents high-level summary metrics including record count, average sales price. Provides stakeholders with a foundational snapshot of the dataset scale and key indicators.

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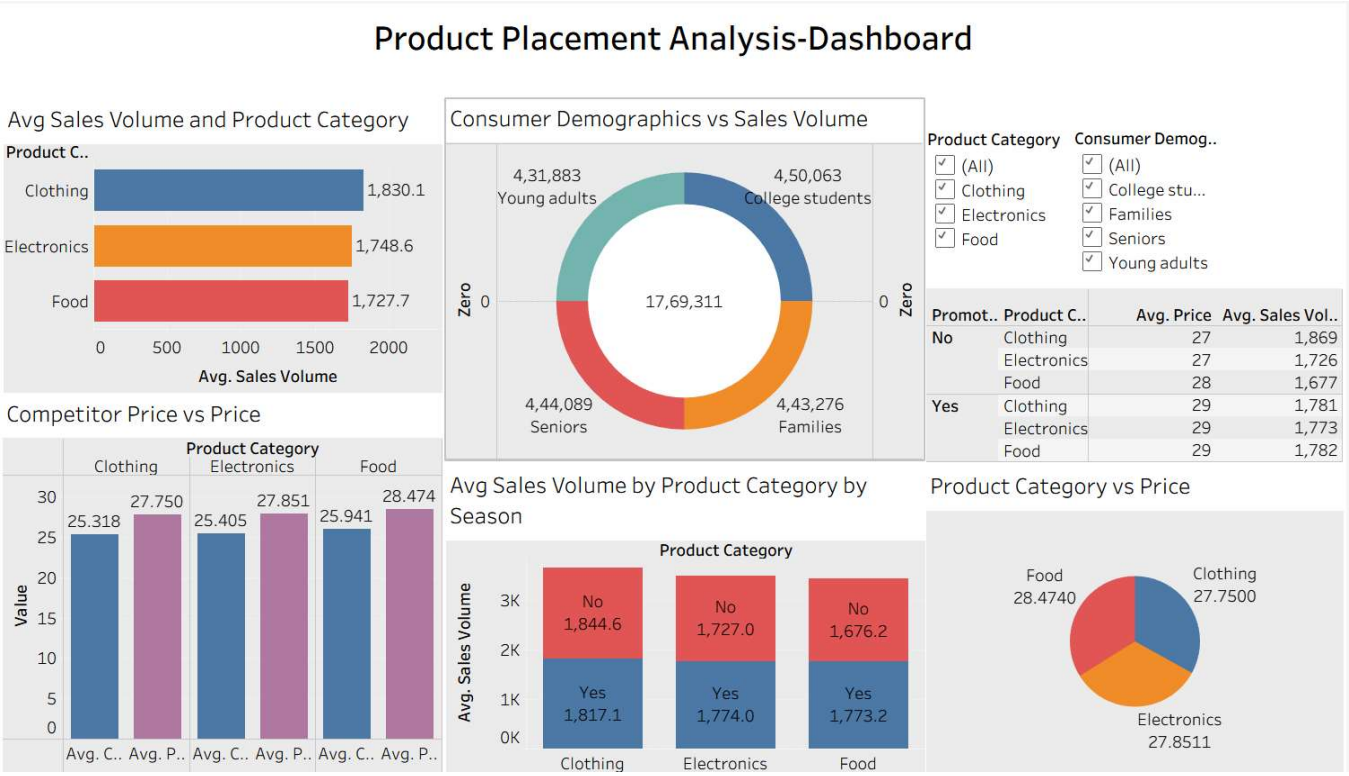
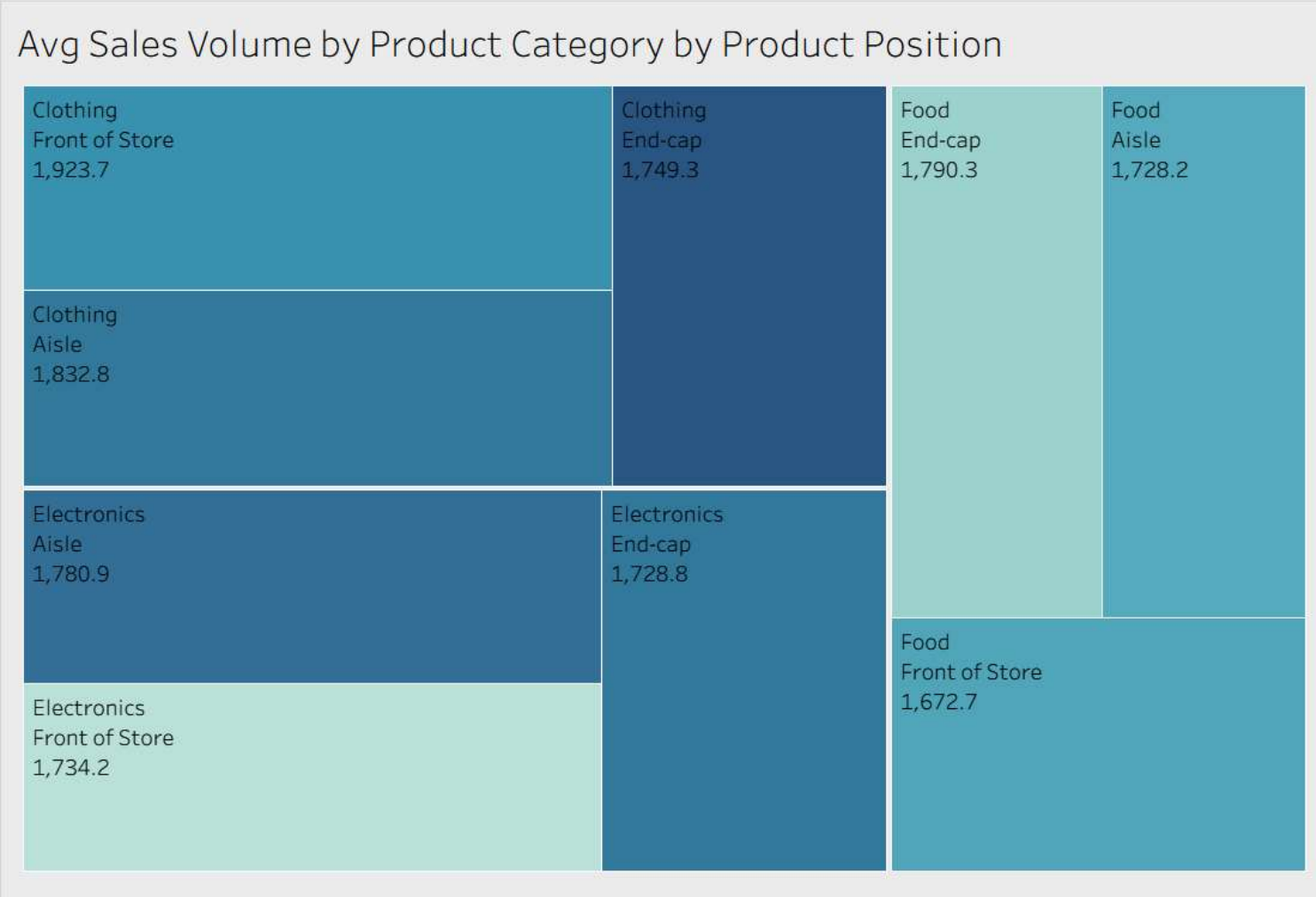


Scenario 2: Avg Sales Volume by Product Category by Product Position

Visualization Type: Tree Map

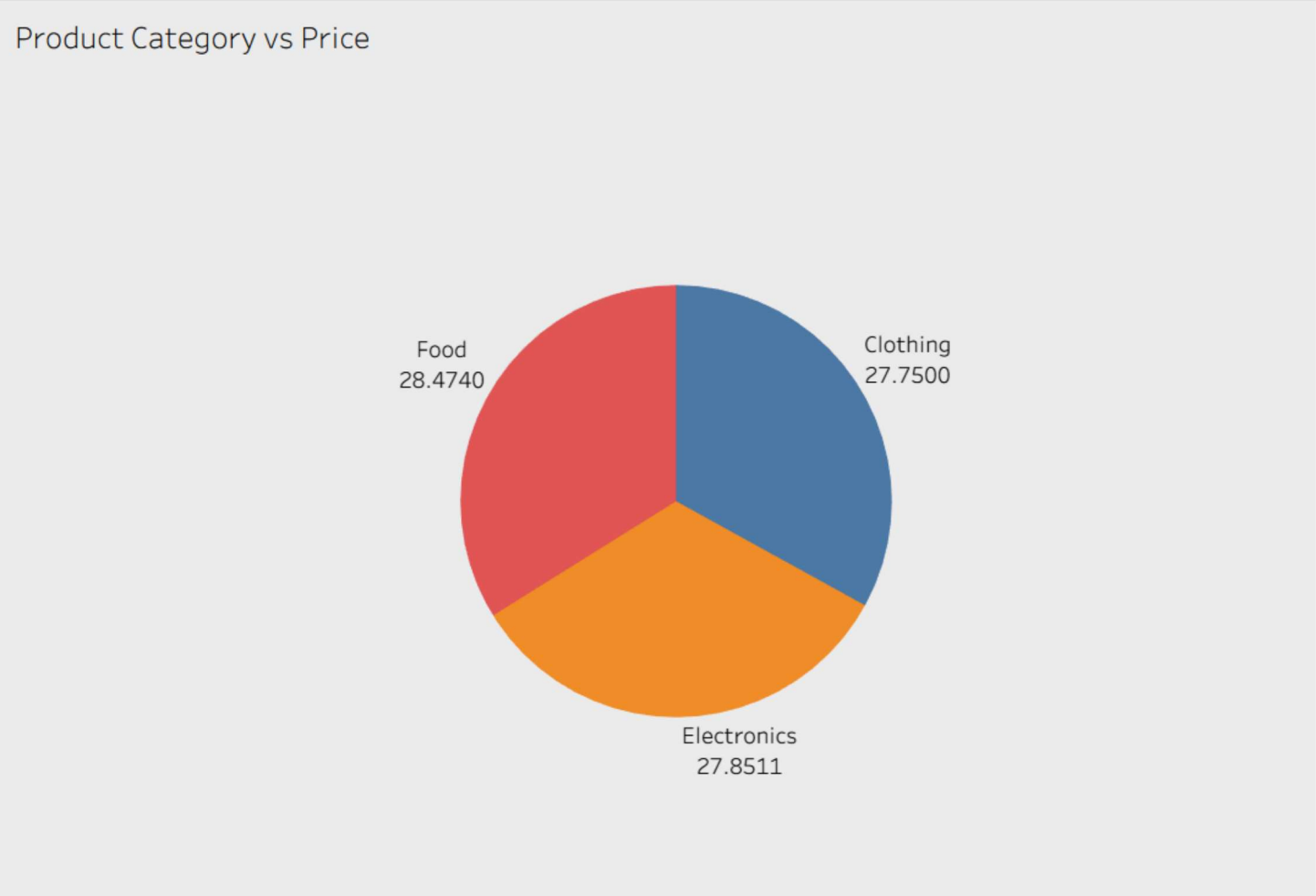
Illustrates the distribution of total sales against product category and position.



Scenario 3: Product Category vs Price

Visualization Type: Pie Chart

This chart helps us in understanding the product category with price in total product placement.



6. Project Activities & Deliverables

All planned project activities have been completed and are reflected in the repository. The table below documents the full project lifecycle:

#	Activity / Task	Description	Status
1	**Data Collection**	Collected and sourced the housing dataset for analysis.	**Completed**
2	**Connect Data with Tableau**	Established data source connection within the Tableau workbook.	**Completed**
3	**Data Preparation**	Cleaned, transformed, and prepared data for accurate visualization.	**Completed**
4	**Data Visualizations**	Created all individual charts and graphs in Tableau.	**Completed**
5	**Dashboard Creation**	Assembled visualizations into interactive Tableau dashboards.	**Completed**
6	**Dashboard Responsiveness**	Configured responsive layouts for different screen sizes.	**Completed**
7	**Storyboard Creation**	Built a Tableau story presenting insights in a narrative sequence.	**Completed**
8	**Performance Testing – Data Filters**	Validated data filter behaviour and cross-view filtering.	**Completed**
9	**Performance Testing – Calculated Fields**	Verified accuracy of all calculated fields in Tableau.	**Completed**
10	**Performance Testing – Visualizations**	Confirmed completeness and correctness of all graphs.	**Completed**
11	**Web Integration (Publishing)**	Published dashboards and storyboard to Tableau Public.	**Completed**
12	**Backend Development (Flask)**	Developed Python/Flask backend with API routes in backend/ folder.	**Completed**
13	**Frontend Development (JS/CSS)**	Built interactive frontend with JavaScript and CSS in frontend/ folder.	**Completed**
14	**Frontend–Backend Integration**	Connected frontend UI to Flask backend and embedded Tableau views.	**Completed**

7. Web Integration

7.1 Backend (Flask — Python)

The backend/ directory contains the Flask web application responsible for:

- Defining URL routes and serving responses to frontend requests
- Providing API endpoints that supply data or configuration to the frontend
- Rendering or directing Jinja2 HTML templates
- Handling any server-side processing required before data is displayed

7.2 Frontend (JavaScript + CSS + HTML)

The frontend/ directory houses the complete client-side codebase:

- JavaScript (40.9%): Manages Tableau embed logic, event handling, dynamic content loading, and user interaction flows.
- CSS (54.7%): Provides all visual styling including layout, typography, colour theming, responsive breakpoints, and component design.
- HTML (1.5%): Base page templates providing document structure, rendered by Flask or served statically.

7.3 Tableau Public Embedding

The Tableau dashboards and storyboard are published to Tableau Public and embedded within the frontend pages via JavaScript embed code. This approach allows:

- Live, interactive visualizations accessible from any modern browser
- No Tableau Desktop licence required for end users
- Seamless integration within the ABC Company branded web interface

8. Performance Testing

Data Filters	All interactive filters tested for correct data segmentation and cross-view filter propagation.
Calculated Fields	All calculated fields reviewed for formula accuracy — including house age, renovation year derivation, and price binning logic.
Visualizations / Graphs	All four visualization scenarios verified for correctness, visual consistency, and completeness across dashboard and storyboard views.

9. Project Conclusion

The Strategic Product Placement Analysis project has been successfully completed. The project has

been elevated with a professional frontend/backend architecture, an enriched JavaScript-driven UI, and comprehensive CSS styling — all version-controlled in the updated GitHub repository.

Key deliverables achieved:

- 4 Tableau visualization scenarios covering sales trends, renovation impact, and feature analysis.
- Interactive Tableau dashboards and storyboard published to Tableau Public.
- Modular Flask backend (backend/) for server-side routing and data serving.
- Rich JavaScript + CSS frontend (frontend/) for an engaging, responsive user experience.
- Full integration of Tableau embeds within the custom web application.

The complete project is version-controlled and available at:

https://github.com/AbhinavSaxena-spec/Strategic-Product-Placement-Analysis-Unveiling-Sales-Impact-with-Tableau-Visualization.git